NARSIMHA REDDY ENGINEERING COLLEGE
PERMANENTLY AFFILIATED TO JNTUH, HYDERABAD - APPROVED BY AICTE, NEW DELHI
AN ISO 9001 : 2008 CERTIFIED INSTITUTE

Project based learning

On

Analysis of Simply Supported beam Using STAAD.Pro Software

BACHELOR OF TECHNOLOGY

IN

DEPARTMENT OF CIVIL ENGINEERING

(2024-2028)

Submitted by

BOLLU PRAVEEN

(24X01A0105)

Introduction

- In this project, STAAD.Pro software is used to model and analyse a simply supported beam. The beam geometry, material properties, cross-sectional dimensions, support conditions, and applied loads are defined in the software. After performing the structural analysis, the results are obtained and studied to understand the behaviour of the beam under loading.
- This analysis helps in understanding the basic principles of structural mechanics and provides practical knowledge of computer-aided structural analysis. It also helps engineers design safe, stable, and economical structures.
- STAAD.Pro is a widely used structural analysis and design software that simplifies complex calculations and provides accurate results for various structural systems.

Project Statement

Themain problem isto analyse a simply supported beam subjected to different loading conditions using STAAD.Pro software. Manual calculations of support reactions, shear forces, bending moments, and deflections can be time-consuming. Therefore, STAAD.Pro is used to create a structural model, assign material properties, define support conditions, and apply external loads.

The analysis aims to determine the structural behaviour of the beam under applied loads and evaluate its strength, stability, and deflection.

The software-generated results are compared with theoretical calculations to understand the accuracy and reliability of the structural analysis.

Objectives

1. To model asimply supported beam using STAAD.Pro software.
2. To define the material properties and cross-sectional dimensions of the beam.
3. To assign appropriate support conditions at both ends of the beam.
4. To apply different types of loads, such as point loads and uniformly distributed loads.
5. To calculate support reactions, shear force, and bending moment.
6. To determine the maximum deflection of the beam under applied loads.
7. To analyse the structural behaviour and stability of the beam.
8. To compare STAAD.Pro analysis results with theoretical calculations.

Proposed Methodology

1. Problem Identification:Select a simply supported beam and identify its dimensions andloading conditions.

2. Structural Modelling: Create the beam model in STAAD.Pro by defining the beam length and geometry.

3. Material Properties: Assign suitable material properties such as Young's modulus, Poisson's ratio, and density.

4. Cross-Section Properties: Define the beam's cross-sectional dimensions, such as width and depth.

6. Load Application: Apply the required loads, such as point loads or uniformly distributed loads.

7. Structural Analysis: Run the analysis in STAAD.Pro to calculate support reactions, shear forces, bending moments, and deflections.

8. Result Interpretation: Study the analysis results using diagrams, tables, and graphical outputs

Final Outcome: The proposed methodology helps evaluate the structural behaviour of a simply supported beam and understand its response under different loading conditions

Tools and technologies

1) STAAD.ProSoftware: Usedforstructural modelling, analysis, and obtaining beam analysis results.

2. Microsoft Excel: Used forcalculations, data recording, and comparison of theoretical and software results.

3. Structural Modelling: Usedtocreate the simply supported beam with appropriate geometry and dimensions.

4. Material Properties: Concreteorsteel properties are assigned based on the selected beam material.

5. Loading Conditions: Pointloadsand uniformly distributed loads are applied to analyse the beam behaviour.

6. Support Conditions: Pin and roller supports are provided to represent a simply supported beam.

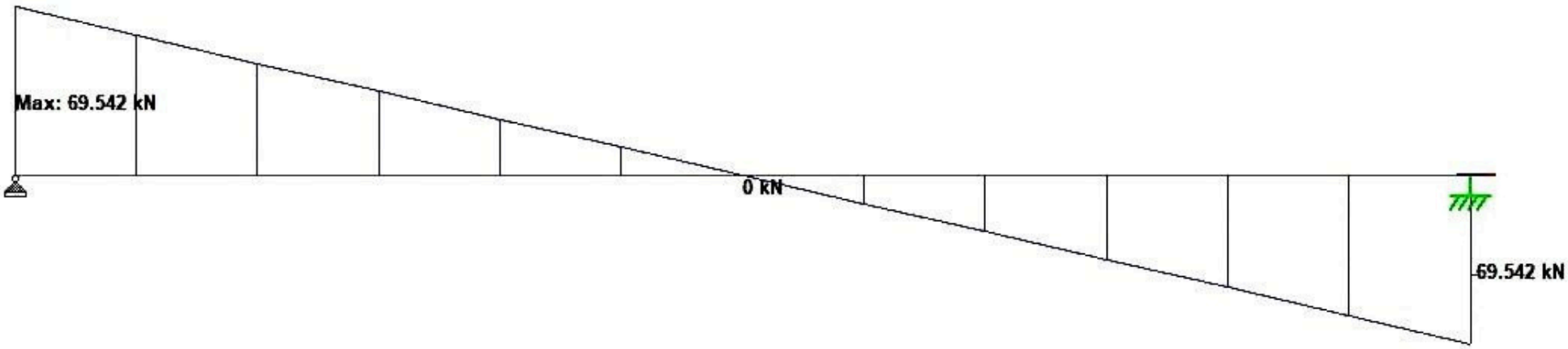
7. Analysis Results: Support reactions, shear force diagrams (SFD), bending moment diagrams (BMD), and deflection values are obtained.

Software Used: STAAD.Pro (Structural Analysis and Design Software).

Results

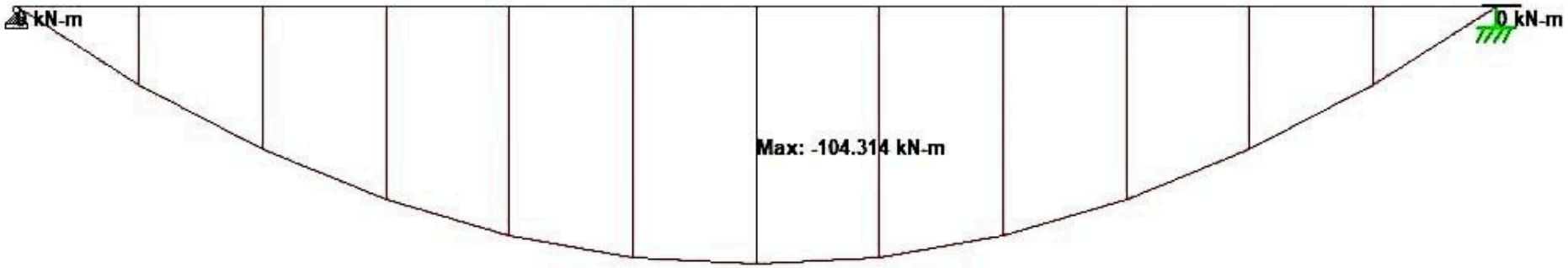
- The simply supported beam was successfully modelled and analysed using STAAD.Pro software. The structural analysis provided the following results:
- The support reactions at both ends of the beam were deter mined.
- Shear Force Diagram (SFD) was obtained to understand the variation of shear force along the beam.
- Bending Moment Diagram (BMD) was generated to identify the maximum bending moment.
- The maximum deflection of the beam under the applied loading conditions was calculated.
- The stress distribution and structural behaviour of the beam were studied.

Shear force diagram



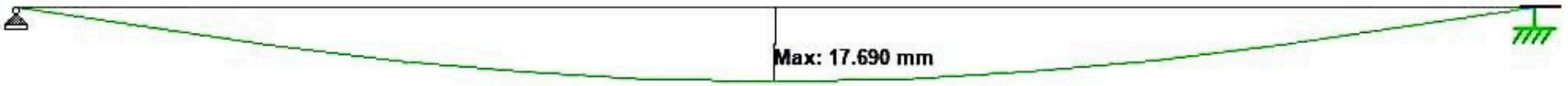
Load 1 : Shear Y
Force - kN

Bending moment diagram



Load 1 : Bending Z
Moment - kN-m

Deflection diagram



Load 1 : Displacement
Displacement - mm

Reference

1.BentleySystems, STAAD.Pro – Structural Analysis and Design Software, Official Website. https://www.bentley.com/software/staad/

2. IS 456:2000, Plain and Reinforced Concrete – Code of Practice, Bureau of Indian Standards, New Delhi.

3. IS 800:2007, General Construction in Steel – Code of Practice, Bureau of Indian Standards, New Delhi.

4. S. Ramamrutham, Strength of Materials, Dhanpat Rai Publishing Company.

5. R. K. Bansal, A Textbook of Strength of Materials, Laxmi Publications.

6. STAAD.Pro Help Documentation, Structural Modelling, Loading and Analysis Procedures, Bentley Systems..

Conclusion

Theanalysis ofa simply supported beam using STAAD.Pro software was successfully carried out. The beam was modelled by defining its geometry, material properties, cross-sectional dimensions, support conditions, and applied loads. The analysis helped determine support reactions, shear forces, bending moments, and maximum deflection of the beam. The results provide a clear understanding of the structural behaviour of the beam under different loading conditions.

STAAD.Pro makes structural analysis easier, faster, and more efficient compared to manual calculations. This project provides practical knowledge of structural modelling and analysis and helps in understanding the application of software in civil engineering.


Thank
You